

PERSONAL LEARNING JOURNEY

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Assigned role: Feature Engineering and Preprocessing

My contribution

My job was to turn the raw data into the inputs (features) that the model learns from. I also had to make sure that none of them uses information from after kickoff. For each team, I built features for its recent form over the last five matches, the strength of its starting eleven and of its three best players. I also measured the strength of its attack, midfield, defence and goalkeeper. I also compared each team's attack with the other team's defence. I grouped the features into four sets, from simple to detailed, so that we could test whether each step helps. I kept a log of every feature, every preprocessing step and every leakage check.

What I learned

I learned how easily leakage can enter a model. Leakage means the model is accidentally given information about the result it is trying to predict. For example, if a team's average form includes the current match, the model already knows part of the answer. The fix is to use only the matches played before it. I also learned that filling in missing values can leak too, if it uses the test data. Putting every step into one scikit-learn Pipeline made sure that these steps learn only from the training data.

Challenge

The two sources measure player quality in different ways. The older data uses FIFA ratings, and the newer data uses market values in euros, which rise every year as transfer prices go up. I needed one fair measure for both. Every player's value also had to be dated before the match. Player positions were recorded differently as well: one source gives positions on the pitch as coordinates, and the other gives position names.

Solution

I turned every player's rating or market value into a percentile from 0 to 100. A percentile shows how a player ranks against earlier players in the same league and position. This puts both sources on the same scale. Only values from before the match were used, and none older than two years (730 days). A team's strength was calculated only when at least 8 of its 11 starters had a value. To prove there is no leakage, I recalculated the form of 1,000 random team-matches from scratch, and all 1,000 matched. All 1,366,594 player values used are dated before their match.

Reflection

The percentile idea worked: on the matches found in both sources, the two measures agree closely (correlation 0.94). I also learned that more features do not always give a better model. Adding the starting eleven clearly helped, adding positions helped a little, and adding formation and line-up changes did not help at all. Proving there is no leakage needs a real test, not only a careful reading of the code.

Future improvement

I would give recent matches more weight than older ones when measuring form, and use minutes played to judge player form better. I would also turn my leakage checks into automatic tests that run every time the dataset is rebuilt.